

2025/26 CSC 4792: Data Mining and Warehousing

Mini Project Practical Assignment

September 4, 2026

1. Assignment Task

Your task is to work in your self-organised groups of five to build a comprehensive dataset containing important information associated with specific City, Municipal, or Town Councils in Zambia. You will extract, clean, and curate this data from the respective council websites as catalogued by the Ministry of Local Government and Rural Development (Republic of Zambia, n.d.).

Please Note: Groups have been pre-assigned specific councils. You must refer to the **Appendices** section at the end of this document to find your group's assigned council and its corresponding official URL. Furthermore, each project team is explicitly required to confirm and specify their assigned topic/council by updating the project teams spreadsheet¹.

1.1. Context: The Role of Local Councils in Zambia

Local authorities in Zambia play a pivotal role in the actualisation of decentralisation and local governance. Their mandates, functions, and financial frameworks are primarily governed by the **Local Government Act No. 2 of 2019** (*Local Government Act, 2019*, 2019), which established an integrated local government system to ensure the effective delivery of services (health, agriculture, infrastructure) at the district level.

Recent legislative amendments have further empowered and restructured the operational frameworks of these authorities:

- **Local Government (Amendment) Act, 2023 (Act No. 28 of 2023)** (*Local Government (Amendment) Act, 2023*, 2023): This amendment reformed the Local Government Equalisation Fund (LGEF), removing prior capital expenditure restrictions and converting it into a recurrent grant. This allows councils to utilise these funds for operational costs, including the timely payment of salaries, thus ensuring stability in local administration.
- **Local Government (Amendment) Act, 2026 (Act No. 76 of 2026)** (*Local Government (Amendment) Act, 2026*, 2026): Enacted alongside massive increments in the

¹ <https://docs.google.com/spreadsheets/d/12IXZd3QSeFUmml2gxbZ1zscZss2gTEIjOf8aZTZugJU/edit?usp=sharing>

Constituency Development Fund (CDF)—now standing at K40 million per constituency in 2026—this Act reinforces the administrative structures, financial accountability, and pension reforms for local authority employees under which councils plan and execute community programmes.

For this group-based mini-project, your dataset must reflect the critical functions and financial footprints of these councils as mandated by the aforementioned Acts. Important data points you must scrape and curate include, but are not limited to:

- **Constituency Development Fund (CDF)** allocations, disbursements, usage, and specific project statuses (e.g., schools, health facilities, water and sanitation).
 - **Approved budgets, LGEF utilisation**, and locally generated revenue streams (e.g., market fees, local taxes, levies).
 - **Integrated Development Plans (IDPs)** and strategic community project records.
 - **Council administrative data**, ward development committee information, and public council resolutions/reports.
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2. Deliverables

This mini-project must be completed strictly within **one week**. The following deliverables must be produced and submitted:

2.1. Dataset on Kaggle

A carefully curated dataset extracted from your assigned council's digital footprints and uploaded to Kaggle.

- **Exemplar Case Study:** To understand the expected standard, structure, and documentation of a high-quality dataset, you **MUST** refer to the exemplar Kaggle dataset used as a case study in class: [A Multi-source Dataset for CS1 Failure Prediction](#) (Phiri, 2026).
- The dataset files must be in **CSV** format.
- You **MUST** use the pipe character "|" as the column separator.
- The data files **MUST** use the naming convention: *db-unza26-csc4792-[DESCRIPTION].csv* (e.g., *db-unza26-csc4792-lusaka_city_council_cdf_projects.csv*).

2.2. Jupyter Notebook and Codebase

A fully functional codebase containing the scripts used for web scraping, data extraction, data cleaning, and preprocessing.

- The notebook **MUST** be committed to a GitHub repository with descriptive, standardised commit messages.
- It must also be shared as a standard Jupyter Notebook file (.ipynb).
- Ensure all steps involved in the creation of the dataset are rigorously documented within the notebook using Markdown cells.

2.3. Data Description Paper

A dataset paper clearly outlining the methodology of how the dataset was created, cleaned, and structured.

- This document **MUST** be written using the "Data in Brief" journal template².
 - The data description paper **MUST** be formatted exactly as outlined in the Data in Brief guide for authors³.
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3. Grading (100 Marks Total)

The Mini Project marks will be distributed as follows:

3.1. Dataset on Kaggle [40 Marks]

- Comprehensiveness, richness, and relevance of the scraped data.
- Strict adherence to the formatting rules (CSV, pipe | separated, and exact naming convention).
- Proper documentation of the dataset on the Kaggle platform (referencing the exemplar model).

3.2. Jupyter Notebook [40 Marks]

- Quality, efficiency, and correctness of data extraction/scraping scripts.
- Data cleaning and preprocessing steps.
- Proper version control (GitHub) and in-code Markdown explanations.

3.3. Data Description Paper [20 Marks]

- Adherence to the "Data in Brief" template and author guidelines.
 - Clarity in explaining the data collection methods, data structure, and the potential reuse value of the dataset.
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² <https://www.elsevier.com/dib-template>

³ <https://www.sciencedirect.com/journal/data-in-brief/publish/guide-for-authors>

4. Submission Instructions

- **Deadline:** All deliverables MUST be submitted no later than **September 12, 2026, 23H59 GMT+2**. No late hand-ins will be accepted.
 - Submit the link to your Kaggle dataset, the link to your GitHub repository (ensure you add the email address lighton.phiri@gmail.com to your repository), and the *.ipynb* file via The Moodle.
 - Submit an electronic copy of your Data Description Paper, in PDF format, via The Moodle.
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5. Bibliography

Local Government Act, 2019. (2019, April 11).

<https://zambialii.org/akn/zm/act/2019/2/eng@2019-04-11>

Local Government (Amendment) Act, 2023. (2023, December 26).

<https://zambialii.org/akn/zm/act/2023/28/eng@2023-12-26>

Local Government (Amendment) Act, 2026. (2026, June 12).

<https://zambialii.org/akn/zm/act/2026/76/eng@2026-06-12>

Phiri, L. (2026). *A Multi-Source Dataset for CS1 Failure Prediction* [Dataset].

<https://www.kaggle.com/lightonphiri/a-multisource-dataset-for-cs1-failure-prediction>

Republic of Zambia. (n.d.). *Ministry of Local Government and Rural Development*. Ministry of

Local Government and Rural Development. Retrieved September 7, 2026, from

<https://www.mlgrd.gov.zm>

Appendices

Appendix A. Pre-Assigned Group Topics and Council URLs

Groups are pre-assigned to the following councils. Please navigate to your respective council's URL to begin your data extraction process. Ensure you update the class Google Spreadsheet with your assigned group and topic.

Table 1: Project Team Topic/Problem Allocation: Republic of Zambia City, Town and Municipal Councils

Project Team #	Name of Council	Website URL
Project Team #1	Lusaka City Council	https://www.lcc.gov.zm
Project Team #2	Ndola City Council	https://www.ndolacitycouncil.gov.zm
Project Team #3	Kitwe City Council	https://www.kcc.gov.zm
Project Team #4	Livingstone City Council	https://www.livingstonecouncil.gov.zm
Project Team #5	Chipata City Council	https://www.chipatacouncil.gov.zm
Project Team #6	Kabwe Municipal Council	https://www.kabwecouncil.gov.zm
Project Team #7	Chingola Municipal Council	https://www.chingolacouncil.gov.zm
Project Team #8	Mufulira Municipal Council	https://www.mufuliracouncil.gov.zm

Project Team #	Name of Council	Website URL
Project Team #9	Luanshya Municipal Council	https://www.luanshyacouncil.gov.zm
Project Team #10	Kalulushi Municipal Council	https://www.kalulushicouncil.gov.zm
Project Team #11	Chililabombwe Municipal Council	https://www.chililabombwecouncil.gov.zm
Project Team #12	Solwezi Municipal Council	https://www.solwezicouncil.gov.zm
Project Team #13	Kasama Municipal Council	https://www.kasamacouncil.gov.zm
Project Team #14	Mansa Municipal Council	https://www.mansacouncil.gov.zm
Project Team #15	Choma Municipal Council	https://www.chomacouncil.gov.zm
Project Team #16	Mazabuka Municipal Council	https://www.mazabukacouncil.gov.zm

Project Team #	Name of Council	Website URL
Project Team #17	Mbala Municipal Council	https://www.mbalacouncil.gov.zm
Project Team #18	Mongu Municipal Council	https://www.mongucouncil.gov.zm
Project Team #19	Chongwe Municipal Council	https://www.chongwecouncil.gov.zm
Project Team #20	Chinsali Municipal Council	https://www.chinsalicouncil.gov.zm
Project Team #21	Kapiri Mposhi Town Council	https://www.kapirimposhiccouncil.gov.zm
Project Team #22	Mumbwa Town Council	https://www.mumbwacouncil.gov.zm
Project Team #23	Serenje Town Council	https://www.serenjecouncil.gov.zm
Project Team #24	Chisamba Town Council	https://www.chisambacouncil.gov.zm

Project Team #	Name of Council	Website URL
Project Team #25	Shibuyunji Town Council	https://www.shibuyunjicouncil.gov.zm
Project Team #26	Mpongwe Town Council	https://www.mpongwecouncil.gov.zm
Project Team #27	Lufwanyama Town Council	https://www.lufwanyamacouncil.gov.zm
Project Team #28	Masaiti Town Council	https://www.masaiticouncil.gov.zm
Project Team #29	Chama Town Council	https://www.chamacouncil.gov.zm
Project Team #30	Lusangazi Town Council	https://www.lusangazicouncil.gov.zm
Project Team #31	Petauke Town Council	https://www.petaukecouncil.gov.zm
Project Team #32	Lundazi Town Council	https://www.lundazicouncil.gov.zm

Project Team #	Name of Council	Website URL
Project Team #33	Mwense Town Council	https://www.mwensecouncil.gov.zm
Project Team #34	Nchelenge Town Council	https://www.nchelengecouncil.gov.zm
Project Team #35	Kawambwa Town Council	https://www.kawambwacouncil.gov.zm
Project Team #36	Samfya Town Council	https://www.samfyacouncil.gov.zm
Project Team #37	Rufunsa Town Council	https://www.rufunsacouncil.gov.zm
Project Team #38	Chilanga Town Council	https://www.chilangacouncil.gov.zm
Project Team #39	Chirundu Town Council	https://www.chirunducouncil.gov.zm
Project Team #40	Kafue Town Council	https://www.kafuecouncil.gov.zm

Project Team #	Name of Council	Website URL
Project Team #41	Monze Town Council	https://www.monzecouncil.gov.zm
Project Team #42	Pemba Town Council	https://www.pembacouncil.gov.zm
Project Team #43	Gwembe Town Council	https://www.gwembecouncil.gov.zm
Project Team #44	Kalomo Town Council	https://www.kalomocouncil.gov.zm
Project Team #45	Zimba Town Council	https://www.zimbacouncil.gov.zm
Project Team #46	Kazungula Town Council	https://www.kazungulacouncil.gov.zm
Project Team #47	Sinazongwe Town Council	https://www.sinazongwecouncil.gov.zm
Project Team #48	Siavonga Town Council	https://www.siavongacouncil.gov.zm

Project Team #	Name of Council	Website URL
Project Team #49	Namwala Town Council	https://www.namwalacouncil.gov.zm
Project Team #50	Itezhi-Tezhi Town Council	https://www.itezhi-tezhicouncil.gov.zm